

Chaitanya Aggarwal

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🌐 [LinkedIn](#)

🐙 [GitHub](#)

Education

Sardar Vallabhbhai National Institute of Technology

B.Tech. in Computer Science & Engineering, CGPA: 8.61

Aug 2024 – Present

Surat, Gujarat

Bhavan Vidyalaya, CBSE

Class XII: 95% — Class X: 95%

Chandigarh

Experience

IIT Delhi

Jun, 2025 – July, 2025

Research Intern | Computer Vision, AI/ML, Python

- Built an automated computer vision pipeline to estimate brickwork quantity from 2D architectural floor plans, eliminating manual measurement workflows.
- Applied EDSR-based image restoration and adaptive binarization to enhance low-quality drawings and enable accurate wall detection and contour extraction.
- Implemented thickness-based wall classification (4.5" vs 9"), pixel-to-real-world scale conversion, and validated results to achieve 78% accuracy against ground-truth calculations.

Projects

ProtoVolt | MERN Stack

[GitHub](#)

- Developed a secure, authenticated web application with personal user workspaces enabling drag-and-drop circuit design and simulation.
- Implemented role-based access (student/professor) with classroom management features similar to Google Classroom for assignments and experiments.
- Integrated Gemini API to analyze user-designed circuits, providing intelligent feedback and learning assistance.
- Built a virtual lab system allowing students to perform and learn experiments anytime, anywhere without physical lab dependency.

SVNIT Club Manager | SQL, Node, HTML/CSS

[GitHub](#)

- Developed a SQL-based DBMS learning application with a web UI integrated to the backend, enabling real-time interaction with the database.
- Implemented CRUD operations to manage students, professors, clubs, events, and sponsors, reinforcing practical DBMS concepts.

Technical Skills

Programming Languages

C/C++, HTML, CSS, JavaScript, Python

Frameworks & Libraries

Node.js, Express

Data Analysis

Pandas, Numpy, Matplotlib, OpenCV,

Databases

MongoDB, MySQL

Software Tools

VS Code, Jupyter Notebook, Google Colab, Github, Postman

Achievements

- Secured 1st position in CAN-SAT 2025, an open-for-all event organised by the Physics Club, SVNIT. Designed a can-sized satellite with a team of four. Received a cash prize and certificate for the achievement.
- Achieved a Codeforces rating of 1027 through regular contest participation.

Extra-Curricular

- Secured 3rd Place for my college in the Inter-NIT Lawn Tennis Tournament.
- Successfully hosted VIRASAT 2025 as a member of SPIC MACAY, anchoring an event featuring Padma Bhushan awardee Vidushi Sudha Raghunathan. Gained valuable experience in event hosting, coordination, and cultural engagement.